

Appendix B — UX Interaction Defaults

Status: Binding for P0 implementation (companion to Spec V2 Rev 3, Appendix A)

Date: 2026-10-01

These defaults govern UX interaction behaviour across the interaction model and renderers. They are implementation contracts, not suggestions. They cover *interaction logic only* — layout, styling, and branding remain design territory. Any deviation requires an explicit update to this Appendix.

Each item is tagged with the phase it is binding for: **[P0]** or **[P1]**.

B1. Wall Editing Interaction

- **[P0]** Vertices are selected by clicking within a `vertexPickRadius` (default 0.15 m world-space, screen-clamped to ≥ 20 px). Wall segments are selected by clicking the segment body away from vertices; segment selection shows segment handles.
- **[P0]** During `MOVING_VERTEX`, the candidate polygon is validated live each frame. A candidate that self-intersects or degenerates renders the affected edge(s) **orange** and the vertex sticks at the last valid position along the drag vector (same "nearest valid" rule as furniture drags, A2). It never renders the room invalid after release.
- **[P0]** **Pre-commit impact preview:** during any wall edit, objects that *would* become hard-invalid after the edit are immediately outlined red (dashed) with a badge count in the status bar ("3 objects will need attention"). This is preview-only; per A3 the edit still commits.
- **[P0]** Segment **thickness** is edited numerically in the inspector when a segment is selected, not by dragging the outer face. The interior polygon is never user-draggable via the thickness rendering; the UI must not expose any handle that implies moving the interior face.
- **[P0]** Vertex insertion (`INSERTING_VERTEX`) is initiated by double-click or long-press on a segment midpoint; a new vertex appears at the quantized click position, splitting the segment into two segments (both inheriting the source segment's thickness). Index remapping per A1.
- **[P0]** Vertex deletion: selecting a vertex and pressing Delete removes it only if the resulting polygon remains valid (≥ 3 vertices, no degenerate edges); otherwise rejected with a status-bar message.

B2. Rotation

- **[P0]** 2D rotation: a rotation handle appears above the selected object's bounding box (offset 24 px above top-centre, hit target 44×44). Angular sensitivity maps pointer arc-distance directly (1:1 angle around the pivot), not linear pixel deltas.
- **[P0]** Group-rotate pivot = **bounding-box centre of the selection** (binding; single source of truth for both the interaction layer and the CompositeCommand).
- **[P0]** **Wall-orientation snap:** when an object's centre comes within `snapDistance` of a wall and its rotation is within 10° of the wall-normal orientation, the rotation snaps to the nearest wall-flush orientation (four candidates: facing away from / into the wall, along $\pm$ wall direction). This is a rotation snap candidate with priority between wall (80) and furniture edge (60) — priority 70 — and participates in the standard snap ranking.
- **[P0]** Shift while rotating disables angular snapping (free rotation), then re-quantizes on release.
- **[P1]** 3D rotation gizmo beyond floor-plane rotation.

B3. Resize

- **[P0]** Handles: 4 edge midpoints + 4 corners on the oriented bounding box. Edge handles resize one dimension; corner handles resize both.
- **[P0] Anchor rule:** the opposite edge/corner stays fixed; the centre moves accordingly. Anchor math is computed by the **domain-provided pure helper** `resizeAboutAnchor` (Phase 1 `footprints.ts`) — the interaction layer calls it and never reimplements centre-shift math, eliminating drift between rotated-handle resizing and domain quantization. The command carries the final quantized position + dimensions.
- **[P0]** Live clamping: during a resize that would violate a hard constraint, the handle sticks at the last valid size (visual clamp, same philosophy as A2). On release, the last valid size is committed. No animate-back needed because the object never left a valid state.
- **[P0]** Minimum size: 0.05 m per dimension, enforced in the interaction layer.
- **[P1]** Per-category aspect-ratio guidance (e.g., beds constrain aspect within $\pm 20\%$ via corner handles unless Alt is held).

B4. Multi-Selection Editing Semantics (Whitelist)

[P0] The following properties have defined per-instance semantics and are editable on multi-selection (all others display as unavailable):

Property	Semantic
width / length / height	set absolute value (per A9 all-or-nothing validity)
elevation	set absolute value
rotation	set absolute value (applied about each instance's own centre)
locked	set on all
finish / material	set on all
metadata fields	set on all

- **[P0]** Position is **not** in the whitelist (absolute position on a group is undefined); position edits are available only via group move.
- **[P0]** Any hard-invalid outcome rejects the whole edit and reverts the field (A9).

B5. Invalid-State Communication

- **[P0]** Two visual severities: **hard** = solid red outline + red fill tint; **soft** = amber dashed outline. The distinction must be visible without reading text.
- **[P0]** Violation hierarchy: the status bar shows the **primary violation only** (per the §4 priority order: outside room → door swing → fixture access → physical collision → clearance). Clicking it opens a popover listing all violations for the selection, each naming involved objects (clickable → select).
- **[P0] Nudge-to-valid affordance:** when a committed object is flagged hard-invalid, its HUD/status popover offers a "Fix position" action that moves the object to the nearest valid position (the A2 nearest-valid computation, exposed as an explicit command — one history entry).
- **[P0]** During drag, the primary violation message updates live; message changes are throttled to once per 100 ms to avoid flicker.

B6. Selection Rules

- **[P0]** Marquee select: **full containment** required (an object is selected only if its entire footprint is inside the marquee rectangle).

- **[P0]** 3D depth picking: raycast returns all hits; the cycle rule is — first click selects the nearest hit; subsequent clicks on the same screen point (within a 500 ms double-click window extended to triple) cycle to the next hit deeper along the ray.
- **[P0]** Selection survives layer hiding and camera occlusion; a selected-but-invisible object shows in the inspector with a "hidden" indicator but remains selected.
- **[P0]** Dashed (above cut-plane) objects are hit-testable in 2D; they participate in picking above floor-level objects at the same screen position using the same cycle rule as 3D depth picking.

B7. Mode & Cancel Hatches

- **[P0]** Escape semantics (binding): `PLACEMENT_GHOST` / duplicate ghost → discard, no history. `DRAGGING` / `RESIZING` / `ROTATING` → restore last committed transform, no history. `MOVING_VERTEX` / `INSERTING_VERTEX` → revert the edit, exit sub-state. `MARQUEE_SELECT` → clear marquee, keep current selection. `MEASURE` → clear current measurement, stay in tool. `WALL_EDIT` → deselect current vertex/segment, stay in tool.
- **[P0]** Persistent tools are exited only by selecting another tool (or a keyboard shortcut 1–4). Escape never exits a persistent tool.
- **[P0]** Switching tools mid-drag cancels the drag per B7 before activating the new tool.

B8. Empty State / First Run

- **[P0]** Empty canvas shows a centred prompt with two actions: "Draw a room" (activates `WALL_EDIT` with inline 3-step hint: click to place first vertex → click to continue → click first vertex / Enter to close) and "Start from a rectangle" (creates a default 4 × 5 m room as one `EditWall` command — undoable).
- **[P0]** After the first room closes, a one-time hint points to the furniture library.
- **[P0]** Tool buttons carry tooltips with their keyboard shortcuts. No further onboarding chrome in P0.

B9. History Visibility

- **[P0]** Undo/redo buttons in the toolbar show the action label of the next entry ("Undo · Move Sofa", "Redo · Place Door"), truncated to fit. Tooltips show the full label.
- **[P0]** `Ctrl/Cmd+Z` / `Shift+Ctrl/Cmd+Z` use the same entries; after any undo the status bar confirms the restored action ("Undid: Move Sofa").
- **[P1]** Full history panel.

B10. Live Dimensions Budget

- **[P0]** During a drag, at most **6** dimension lines are shown: nearest wall (per dominant axis, max 2), nearest furniture edge (max 1 per axis, 2 total), nearest fixture (1), alignment axis indicator (1). Nearest = smallest perpendicular distance.
- **[P0]** Dimension labels are never drawn overlapping each other; on collision the lower-priority label (order: wall > fixture > furniture > alignment) is offset along its dimension line by one label height, and suppressed if no free position exists within 2 label heights.
- **[P0]** Below screen-scale 0.15 (world metres per screen pixel), only wall dimensions are shown; the rest are suppressed.
- **[P0]** Labels render with a solid background chip; text is display-unit formatted per §3 Units.